

Lista 4 - Redes de Computadores I

Nome: Thiago Henrique Gomes Feliciano

1) Crie uma aplicação (Chat) UDP em Java usando os códigos de UDP vistos em sala de aula.

R:

Cliente.java:

```
J Servidor.java 2  J Cliente.java 3 X  J Cliente.class  J Servidor.class
J Cliente.java > Language Support for Java(TM) by Red Hat > Cliente > lerString()
1  import java.io.*;
2  import java.net.*;
3
4  class Cliente {
5      private static int portaServidor = 9871;
6      private static byte[] sendData = new byte[1024];
7      private static byte[] receiveData = new byte[1024];
8
9      public static byte[] lerString() throws Exception {
10         BufferedReader in = new BufferedReader(new InputStreamReader(System.in));
11         return in.readLine().getBytes();
12     }
13
14     public static void main(String args[]) throws Exception {
15         DatagramSocket clientSocket = new DatagramSocket();
16
17         InetAddress ipServidor = InetAddress.getByName(host: host: "192.168.255.10"); // ip do servidor
18
19         sendData = lerString();
20
21         DatagramPacket sendPacket = new DatagramPacket(sendData, sendData.length, ipServidor, portaServidor);
22
23         clientSocket.send(sendPacket);
24
25         DatagramPacket receivePacket = new DatagramPacket(receiveData, receiveData.length);
26
27         clientSocket.receive(receivePacket);
28
29         clientSocket.close();
30
31         System.out.println("FROM SERVER:" + new String(receivePacket.getData()));
32     }
}
```

Servidor.java:

```
Servidor.java 2 X  Cliente.java 3  Cliente.class  Servidor.class X
J Servidor.java > Language Support for Java(TM) by Red Hat > Servidor
1  import java.net.*;
2
3  class Servidor {
4      private static int portaServidor = 9871;
5      private static byte[] receiveData = new byte[1024];
6      private static byte[] sendData = new byte[1024];
7
8      public static void main(String args[]) throws Exception {
9          try (DatagramSocket serverSocket = new DatagramSocket(portaServidor)) {
10              while (true) {
11                  DatagramPacket receivePacket = new DatagramPacket(receiveData, receiveData.length);
12
13                  System.out.println(x: "Aguardando datagrama do cliente...");
14                  serverSocket.receive(receivePacket);
15
16                  System.out.println("RECEIVED: " + new String(receivePacket.getData()));
17
18                  InetAddress ipCliente = receivePacket.getAddress();
19                  int portaCliente = receivePacket.getPort();
20
21                  sendData = (new String(receivePacket.getData())).toUpperCase().getBytes();
22
23                  DatagramPacket sendPacket = new DatagramPacket(sendData, sendData.length, ipCliente, portaCliente);
24                  serverSocket.send(sendPacket);
25
26                  System.out.println(x: "Enviado...");
27              }
28          }
29      }
30  }
```

Aplicação (Chat) UDP:

```
PS C:\Users\thigo\Downloads\Lista-redes> javac Servidor.java
PS C:\Users\thigo\Downloads\Lista-redes> javac Cliente.java
PS C:\Users\thigo\Downloads\Lista-redes> java Servidor
```

```
Aguardando datagrama do cliente...
RECEIVED: oi, boa tarde
Enviado...
Aguardando datagrama do cliente...
█
```

```
PS C:\Users\thigo\Downloads\Lista-redes> javac Cliente.java
PS C:\Users\thigo\Downloads\Lista-redes> java Cliente
oi, boa tarde
FROM SERVER:OI, BOA TARDE
PS C:\Users\thigo\Downloads\Lista-redes> █
```